

Project Charter - Group 2

Boilertronics

Team Members:

Keenan Koehler, Ethen Chen, Ethan Dixon, Zachary Kirkeby, Abhi Doshi

Project Title:

Boilertronics

Problem Statement:

Many games are made just for you to relax and have fun. But, as computer science students, our fun is through solving problems while taking into account many factors. In this game you must optimize a factory for multiple different metrics such as time efficiency, total cost, and profit per cycle. Taking inspiration from games made by Zachtronics, but with our own programming language, mechanics, and story. Our main goal with this game is to allow for people to both relax and have fun while staying sharp and keeping their brain engaged.

Project Objectives:

- Create an engaging game that challenges the player via factory building/programming levels
- Create a system to 'grade' the user's solution on a set list of criteria
- Create a scripting language to allow the user to program their factories
- Create a set of objects/machines/tracks in a 2D environment that the user uses to build a solution to the level
- Allow the user to run and see their solution work in real time while detecting errors with the solution
- Allow the user to navigate to different levels via a level menu
- Create different sections in the level menu that will increase in difficulty and challenge the user on different aspects of the game
- Allow the user to save, load, and delete solutions
- Create different obstacles in the levels that the user will have to overcome
- Allow the user to set different 'tiers' of machines they use to change the quality of the output
- Develop tutorials and documentation on how different mechanisms of the game works so that the user can reference them

Stakeholders:

- **Customers/Users:** People who play the game, wanting to test their spatial reasoning and programming logic
- **Developers:** Keenan Koehler, Ethen Chen, Ethan Dixon, Zachary Kirkeby, Abhi Doshi
- **Project Coordinator:** Bishal Papan

- **Project Owners:** Keenan Koehler, Ethen Chen, Ethan Dixon, Zachary Kirkeby, Abhi Doshi

Project Deliverables:

- A 2D factory building/programming game developed in C# using the Godot engine for personal computers
 - A system that grades a user's solution based on set criteria: time efficiency, cost, profit per cycle and has a leaderboard type system
 - A scripting language built into the game that allows the user to program 'grabbers' and interact with the factory
 - A set of objects, machines, and tracks that the user can place and interact with via their scripts
 - A system for the user to be able to step through, run, pause, and reset their solution
 - A level navigation window that allows the user to navigate through different levels
 - Sets of unique levels that will challenge users on different mechanics of the game
 - A scripting language built into the game based on a reduced instruction subset of a generic, assembly style language
- A file format and parsing system that allows the user to save and load different levels/solutions into their game
- A set of tutorials and documentation to help the user to learn the different systems and mechanics of the game

Tech Stack:

- C#
- Custom ASM instruction set
- Godot